University of Dhaka

Department of Applied Mathematics

Second Year B S (Honours) 2025-2026

Course Title: Math Lab II   Course Code: AMTH 250

Assignment 01

Name: **Roll:** **Group:**

[Write a Fortran Program to solve each of the following problems. Use files for input/output unless specified otherwise. Name the files and the code according to the assignment and problem no., e.g., for problem no. Y of Assignment X, input and output file names must be 'inXqY.txt' and 'outXqY.txt' respectively.]

1. Write a **FORTRAN program** which reads the following matrix from a file and stores it as **matrix A**

$$\mathbf{A} = \begin{pmatrix} 4 & 7 & 2 & 5 \\ 1 & 6 & 8 & 3 \\ 9 & 2 & 5 & 7 \\ 3 & 8 & 1 & 4 \end{pmatrix}$$

In the output file:

(i) Write A in matrix from

(ii) Write only the elements whose row number is even.

(iii) Write only the elements of 1st row, and 1st column and 4th row and 4th column as

4	7	2	5
1			3
9			7
3	8	1	4

(iv) Construct a new matrix B by replacing all boundary elements with 0, while keeping all other elements unchanged. Output of matrix B is $\begin{matrix} 0 & 0 & 0 & 0 \\ 0 & 6 & 8 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}$

2. Write a **FORTRAN program** to store the following name in a one-dimensional character array:

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Write programs to perform the following operations:

(i) Print the name as it appears and in reverse order.

(ii) Count and print the total number of uppercase letters, lowercase letters, vowels, consonants, digits, and blank spaces.

(iii) Replace every vowel with the character '*' and print the modified name.

(iv) Determine whether the given character array represents a palindrome (ignoring blank spaces and letter case). Print an appropriate message.

3. Write a **FORTRAN program** to generate a **7×7 matrix of random real numbers**. Store this matrix in **three different files** named *a1q3_F txt, a1q3_E txt, a1q3_ES txt* respectively using
   (i) The F descriptor, corrected up to 10 decimal places
   (ii) The E descriptor, corrected up to 8 decimal places
   (iii) The ES descriptor, corrected up to 7 decimal places
   Find Mean, Maximum and Minimum from the numbers.

4 (i) Write a **FORTRAN program** using a **subroutine** to compute the number of positive divisors$\left(\tau(n)\right)$ and the sum of positive divisors $\left(\sigma(n)\right)$ of a given positive integer.

(ii) A perfect number is a positive integer whose sum of divisors (excluding itself) equals the number itself — equivalently $\sigma(n) = 2n$. Using your **subroutine** from 4(i), write a **FORTRAN program** to find and print the first 4 perfect numbers.

5 The Möbius function is defined as

$$\mu(n) = \begin{cases} 1, & n = 1 \\ (-1)^k, & n \text{ is the product of k distrinct primes} \\ 0, & \text{otherwise} \end{cases}$$

Write a **FORTRAN function t**hat returns the value of the Möbius function for a given positive integer n. (Example: $\mu(30) = -1,\ \mu(12) = 0,\ \mu(6) = 1$).

6 (i) For any two positive integers a and b the GCD satisfies the following recurrence relationship:

$$gcd(a, b) = \begin{cases} b, & a\ (mod\ b) = 0 \\ gcd\big(b, a\ (mod\ b)\big), & a\ (mod\ b) \neq 0 \end{cases},$$

write a **program** using **recursive function** that takes a and b as input and gives their GCD as output.

(ii) Using your previous **recursive function** from 6(i), write a **FORTRAN program** that reads n positive integers into an array and computes the GCD of all n numbers, by repeatedly applying the two-number GCD function across the array.